

Notes: Jansen (RFS, 2025)

Long-Term Investors, Demand Shifts, and Yields

Contents

1	Big picture	3
2	Data and measurement (institutional context and key objects)	4
2.1	Why the Dutch P&I sector is the right laboratory	4
2.2	Regulatory framework before and after the reform	4
2.3	The UFR reform and the regulatory discount curve	4
2.4	Data sources and sample construction	5
2.5	Main sample and descriptive scale	5
2.6	Main quantity variables	6
2.7	Main exposure variables	6
2.8	Yield and demand-system objects	7
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Conceptual mechanism and hypotheses	7
3.2	Baseline long-maturity bond-holdings regression	8
3.3	Risky-asset allocation regression	8
3.4	Constraint-sensitive triple-difference specification	8
3.5	Demand system for Dutch government bonds	9
3.6	Instrumenting for yields	9
3.7	Price elasticities of demand	9
3.8	Allowing substitution with other safe euro-area bonds	10
3.9	Short-run versus long-run elasticities	10
3.10	European external-validity regression	10
4	Identification and empirical design (what makes the estimates credible)	10
4.1	Why the UFR reform is close to exogenous	10
4.2	Why liability duration is the right exposure measure	11
4.3	Why the constraint interaction matters	11
4.4	Why quantity evidence is especially persuasive	11
4.5	Alternative explanations the paper confronts	11
4.6	What the Dutch design can and cannot identify	12
4.7	Relation to the literature	12

5	Tables and figures	12
5.1	Figure 1: Dutch 30-20 year yield spread around the reform	12
5.2	Figure 2: How the UFR changes the regulatory discount curve	13
5.3	Table 1 and Figure 3: Sample structure and portfolio rebalancing over time	13
5.4	Table 2: Long-term bond holdings and the regulatory discount curve	13
5.5	Table 3: Asset allocation and the risky-asset channel	15
5.6	Table 4: Constraints amplify the maturity shift	15
5.7	Table 5 and Table 6: Instrument and demand-system estimates	16
5.8	Table 7 and the 30-year pricing back-of-the-envelope	16
5.9	Figures 4 and 5: Why banks absorb the demand shock	17
5.10	Table 8 and Figure 6: European evidence	18
6	Short summary	18
7	Conclusion	19
7.1	Core conclusion (what the paper establishes)	19
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** How much do long-term-investor demand shifts move long-end government bond yields, and which investors ultimately absorb those shocks?
- **Setting.** The paper studies a Dutch reform to the regulatory discount curve used to value the liabilities of pension funds and insurance companies (P&Is). The reform was announced and implemented on July 2, 2012.
- **Core challenge.** Price data alone cannot tell us how important demand shocks are or who the marginal arbitrageurs are. The paper therefore combines holdings data, liability data, and yield data.
- **Main conceptual contribution.** The reform changed the maturity exposure of P&I liabilities in a very specific way: more exposure to the 20-year rate, less exposure to very long rates. That gives a maturity-specific demand shock that can be traced in both quantities and prices.
- **Mechanism.** P&Is hedge both economic liabilities and regulatory liabilities. Once regulation makes the liability value depend more on the 20-year rate and less on longer rates, optimal hedging shifts out of 30-year bonds and toward bonds near 20 years.
- **Main empirical design.** The paper first estimates difference-in-difference style portfolio regressions across Dutch pension funds and insurers, using pre-reform liability duration and solvency positions to measure exposure. It then uses the reform as an instrument for maturity-bucket yields in an asset-demand system for government bonds.
- **Main quantity result.** Long-duration P&Is reduce holdings of bonds with remaining maturity at least 30 years and increase holdings of bonds with maturities between 15 and 25 years. The aggregate decline in 30-year bond holdings is about EUR 10.15 billion, roughly 33% of pre-reform holdings, while 20-year holdings rise by about EUR 15.71 billion, or 17% of pre-reform holdings.
- **Main price result.** The reform steepens the long end of the Dutch yield curve. In the first stage, yields around 20 years fall while those above 25 years rise; a back-of-the-envelope calculation implies about a 17 basis point increase in 30-year yields in the short run.
- **Key risk-taking result.** P&Is with longer liabilities also increase risky asset exposure after the reform: they move from corporate bonds toward equities and riskier government bonds.
- **Key heterogeneity result.** Constrained institutions react more strongly. P&Is closer to their solvency constraints cut 30-year bond holdings and raise 20-year holdings more aggressively.
- **Key market-mechanism result.** Banks are the most price-elastic investors in Dutch government bond markets and primarily absorb the shock. Their median price elasticity is about 14.95, far above other sectors.
- **External validity result.** Similar, though smaller, steepening effects appear in other safe European bond markets and in a European-country panel once Solvency II introduces the UFR more broadly.

- **Economic takeaway.** Regulation of long-term investors changes the maturity structure of demand for safe assets and therefore changes governments' borrowing costs. The spillovers run directly into banks and other sectors.

2 Data and measurement (institutional context and key objects)

2.1 Why the Dutch P&I sector is the right laboratory

- The paper studies a clean regulatory shock that directly changes how long-term liabilities are discounted.
- The Dutch setting is unusually rich because the regulator observes detailed bond holdings, derivative positions, and liabilities.
- This is economically important, not a niche market. The pension sector alone manages roughly EUR 1.75 trillion, about 192% of Dutch GDP and around 58% of euro-area pension-fund assets.
- P&Is are natural preferred-habitat investors because their liabilities are long dated, and they hedge interest-rate risk with both bonds and swaps.

Interpretation. The setting gives both a source of quasi-experimental variation and the exact balance-sheet objects needed to see how demand changes.

2.2 Regulatory framework before and after the reform

- Since 2007, assets are marked to market for both pension funds and insurance companies.
- A pension fund's key regulatory object is the *funding ratio*, assets divided by liabilities. Required funding is risk-based; if the ratio is too low, the fund must file a recovery plan. Below 104.2%, pension rights cannot be indexed; below 90%, accrued benefits may have to be cut.
- For insurers, the key object is the *solvency ratio*. Before Solvency II, required capital was a simple 4% of liabilities; under Solvency II it becomes risk-based and stricter.

Interpretation. P&Is do not just care about economic asset-liability matching. They also care about their regulatory funding positions, because those positions affect indexation, dividends, and supervisory intervention.

2.3 The UFR reform and the regulatory discount curve

Before July 2012, the Dutch regulator discounted liabilities using market euro swap rates over the full curve, so the regulatory and economic discount curves coincided. The reform replaced the long end with an *ultimate forward rate* (UFR) extrapolation.

For forward rates, the post-reform curve is:

$$f_{t,h-1}^{h,*} = \begin{cases} f_{t,h-1}^h, & 1 \leq h \leq 20, \\ (1 - w_{UFR}(h)) f_{t,19}^{20} + w_{UFR}(h) \cdot UFR, & 20 < h < 60, \\ UFR, & h \geq 60. \end{cases} \quad (1)$$

The corresponding zero-coupon rates satisfy

$$(1 + y_t^{(h)})^h = \prod_{n=1}^h (1 + f_{t,n-1}^{n,*}), \quad h = 1, 2, \dots, 120. \quad (2)$$

Interpretation. The reform has three first-order effects:

- current liability values fall, because the UFR is above market long rates;
- regulatory liabilities become *less* sensitive to parallel long-rate shifts beyond 20 years;
- regulatory liabilities become *more* sensitive to the 20-year rate, because that rate partly anchors all maturities beyond 20 years.

This is the central mapping from regulation to bond demand.

2.4 Data sources and sample construction

The paper combines four data environments:

- **Dutch SHS holdings data** for pension funds and insurance companies, quarterly from 2009q1 to 2019q4, with security-level positions.
- **Supervisory data** on liabilities, funding ratios, asset allocations, and liability durations.
- **CSDB security characteristics** for maturity, coupon, yield to maturity, prices, and amount outstanding.
- **EMIR derivatives data** from 2017 onward, used to study swap positions and counterparties.

For the demand-system section, Dutch data are extended with ECB Securities Holdings Statistics by Sector (SHSS), giving sector-country holdings of Dutch government debt for euro-area investors.

2.5 Main sample and descriptive scale

The institutional sample covers:

- 42 pension funds,
- 12 life insurers,
- 27 nonlife insurers.

Together these institutions account for roughly 80%–90% of Dutch institutional-investor assets under management.

Table 1 shows the main cross-sectional facts:

- Life insurers are the largest on average; pension funds are next.
- Government bonds account for about 49% of life-insurer assets, 45% of nonlife-insurer assets, and 38% of pension-fund assets.
- Corporate bonds are important for insurers, while pension funds hold far more equity.
- Liability duration is longest for pension funds (about 17.9 years), then life insurers (11.8), and shortest for nonlife insurers (4.2).

Interpretation. There is large heterogeneity in how exposed institutions are to the regulatory change, and that heterogeneity is exactly what the empirical design exploits.

2.6 Main quantity variables

The paper works with maturity buckets rather than individual bonds, because neighboring maturities are close substitutes. The key buckets are:

- **30-year bonds:** remaining maturity $T \geq 30$,
- **20-year bonds:** $15 < T \leq 25$,
- and a shorter-maturity bucket $T \leq 15$.

The main dependent variables are portfolio shares in those maturity buckets, measured in notional amounts so that results reflect active quantity choices rather than valuation changes.

2.7 Main exposure variables

The two central exposure measures are:

- **Liability duration** as of 2011q2, $D_{2011q2,i}^L$, which captures how much the reform should matter for institution i .
- **Inverse funding ratio minus one**, $FR_{it}^{-1} - 1$, which captures how constrained the institution is. Values above zero mean underfunding.

Interpretation. Long liability duration tells you who is hit hardest by the new regulatory curve; low solvency tells you who has the strongest incentive to hedge the regulatory rather than economic curve.

2.8 Yield and demand-system objects

For the pricing part of the paper, investor i 's holdings of Dutch government bonds in maturity bucket h are denoted $B_{it}(h)$ and the outside asset is O_{it} . Portfolio weights are defined as

$$w_{it}(h) = \frac{B_{it}(h)}{O_{it} + \sum_{h=1}^H B_{it}(h)} = \frac{\delta_{it}(h)}{1 + \sum_{h=1}^H \delta_{it}(h)}. \quad (3)$$

The outside asset is proxied by the 10-year German yield in the baseline Dutch-only demand system.

Interpretation. The demand system is a maturity-bucket portfolio-allocation model rather than a single-bond regression.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual mechanism and hypotheses

The paper's model is a partial-equilibrium mean-variance framework with assets and liabilities. P&Is care about the volatility of both their *economic* and *regulatory* funding ratios.

Let $F_t^E = A_t/L_t^E$ be the economic funding ratio and $F_t^R = A_t/L_t^R$ the regulatory funding ratio. A simplified objective is:

$$\max_{w_t} E_t \left[\frac{A_{t+1}}{A_t} \right] - \frac{\gamma}{2} Var_t \left[\frac{A_{t+1}}{A_t} - \frac{L_{t+1}^E}{A_t} \right] - \frac{\lambda(F_t^R)}{2} Var_t \left[\frac{A_{t+1}}{A_t} - \frac{L_{t+1}^R}{A_t} \right]. \quad (4)$$

The parameter $\lambda(F_t^R)$ declines with the regulatory funding ratio, so poorly funded institutions care more about regulatory mismatch.

The optimal risky-asset demand is pure speculative demand:

$$w_{S,t}^* = \frac{E_t[r_{t+1}^S - r_f]}{(\gamma + \lambda(F_t^R)) Var_t[r_{t+1}^S]}, \quad (5)$$

while the optimal bond portfolio is the sum of speculative demand, economic hedging demand, and regulatory hedging demand:

$$w_{B,t}^* = \underbrace{\frac{E_t[r_{t+1}^B - r_f \mathbf{1}]}{(\gamma + \lambda(F_t^R)) Var_t[r_{t+1}^B]}}_{\text{speculative}} + \underbrace{\frac{\gamma}{\gamma + \lambda(F_t^R)} \frac{1}{F_t^E} a_t \circ \frac{Cov_t(r_{t+1}^B, r_{t+1}^L)}{Var_t(r_{t+1}^B)}}_{\text{economic hedge}} + \underbrace{\frac{\lambda(F_t^R)}{\gamma + \lambda(F_t^R)} \frac{1}{F_t^R} \left(a_t \circ \xi^L \circ \frac{Cov_t(r_{t+1}^B, r_{t+1}^L)}{Var_t(r_{t+1}^B)} \right)}_{\text{regulatory hedge}} \quad (6)$$

Interpretation. The reform changes ξ^L : it raises the regulatory sensitivity around 20 years and lowers it around 30 years. That mechanically tilts regulatory hedging demand away from 30-year bonds and toward 20-year bonds.

The model yields three empirical predictions:

- P&Is with longer liability duration cut 30-year holdings more and raise 20-year holdings more.
- Those same institutions raise risky asset holdings more because the reform provides larger capital relief.
- More constrained P&Is react more strongly because they place more weight on regulatory hedging.

3.2 Baseline long-maturity bond-holdings regression

The main holdings regression is

$$w_{it}^B = \alpha + \beta_1 D_{2011q2,i}^L \times UFR_t + \beta_2 FR_{it-1}^{-1} + \beta_3 FR_{it-1}^{-1} \times PF_i + \beta_4 D_{it-1}^L + \beta_5 AUM_{it-1} + \beta_6 Vol FR_{it-1}^{-1} + \nu_i + \lambda_t + \varepsilon_{it}. \quad (7)$$

Here w_{it}^B is the fraction of the bond portfolio in a given maturity bucket, UFR_t equals one after 2012q3, and PF_i is a pension-fund dummy.

Interpretation. The interaction $D_{2011q2,i}^L \times UFR_t$ is the treatment effect: longer-duration liabilities should create larger post-reform shifts in duration demand.

3.3 Risky-asset allocation regression

To study the reform's capital-relief channel, the paper estimates

$$w_{it}^S = \alpha + \beta_1 D_{2011q2,i}^L \times UFR_t + \beta_2 FR_{it-1}^{-1} + \beta_3 FR_{it-1}^{-1} \times PF_i + \beta_4 D_{it-1}^L + \beta_5 AUM_{it-1} + \beta_6 Vol FR_{it-1}^{-1} + \nu_i + \lambda_t + \varepsilon_{it}, \quad (8)$$

where w_{it}^S is now an allocation to equities, corporate bonds, government bonds, or riskier subcomponents of those buckets.

Interpretation. If the reform raises solvency mechanically by lowering liability values, exposed P&Is should increase speculative positions.

3.4 Constraint-sensitive triple-difference specification

To test whether constrained institutions react more strongly, the paper estimates

$$\begin{aligned} w_{it}^B = & \alpha + \beta_1 D_{2011q2,i}^L \times FR_{2011q2,i}^{-1} \times UFR_t + \beta_2 D_{2011q2,i}^L \times UFR_t \\ & + \beta_3 FR_{2011q2,i}^{-1} \times UFR_t + \beta_4 FR_{it-1}^{-1} + \beta_5 FR_{it-1}^{-1} \times PF_i \\ & + \beta_6 D_{it-1}^L + \beta_7 AUM_{it-1} + \beta_8 Vol FR_{it-1}^{-1} + \nu_i + \lambda_t + \varepsilon_{it}. \end{aligned} \quad (9)$$

Interpretation. The triple interaction asks whether high-duration, weakly funded institutions shift maturity the most after the reform.

3.5 Demand system for Dutch government bonds

The paper then moves from quantities to prices using the Kojien-Yogo demand system. For investor i and maturity bucket h ,

$$\ln B_{it}(h) = \hat{\alpha}_i + \beta_{0i}y_t(h) + \beta'_{1i}x_t(h) + \hat{\beta}_{2i}y_t^{DE} + \beta_{3i}\ln(B_{2009q1,i}(h)) + \varepsilon_{it}(h), \quad (10)$$

where $y_t(h)$ is the yield in maturity bucket h , $x_t(h)$ contains bond characteristics, and y_t^{DE} is the 10-year German yield.

Bond controls include duration, convexity, coupon, and the log of total amount outstanding.

Interpretation. This is a sector-level demand curve: how much does each investor sector want to hold in a given maturity bucket when yields move?

3.6 Instrumenting for yields

Because latent demand can itself move yields, the paper instruments the maturity-bucket yields with the UFR-implied weights assigned to each bucket:

$$z_t(h) = \xi(h) UFR_t. \quad (11)$$

The first-stage equation is

$$y_t(h) = \gamma_{0i} + \gamma_{1i}z_t(h) + \gamma'_{2i}x_t(h) + \gamma_{3i}y_t^{DE} + \gamma_{4i}\ln(B_{2009q1,i}(h)) + \varepsilon_{it}(h). \quad (12)$$

The instrument values are negative in the 15–20 and 20–25 buckets and positive beyond 25 years, exactly matching excess demand at 20 years and excess supply at 30 years.

Interpretation. The reform creates a maturity-specific relative-price shock, not just a parallel shift in yields.

3.7 Price elasticities of demand

Once β_{0i} is estimated, the demand elasticity with respect to price is computed as

$$-\frac{\partial q_{it}(h)}{\partial p_{it}(h)} = 1 + 100 \frac{\beta_{0i}}{T_t(h)} (1 - w_{it}(h)). \quad (13)$$

Interpretation. Large positive values mean sector i is willing to reallocate strongly across maturity buckets when relative prices change.

3.8 Allowing substitution with other safe euro-area bonds

Because Dutch P&Is also rebalance in German and French bonds, the paper extends the demand system to separate Dutch-specific from safe-euro movements:

$$\ln B_{it}(h) = \hat{\alpha}_i + \beta_{0i} y_t^{NL,S}(h) + \beta_{1i} y_t^{EU}(h) + \beta'_{2i} x_t(h) + \hat{\beta}_{3i} y_t^{US} + \beta_{4i} \ln(B_{2009q1,i}(h)) + \varepsilon_{it}(h). \quad (14)$$

Here $y_t^{NL,S}(h)$ is the Dutch yield spread relative to the safe-euro yield, and $y_t^{EU}(h)$ is the safe-euro benchmark yield.

Interpretation. This is a segmentation check. If Dutch bonds were perfect substitutes for other safe euro bonds, Dutch-specific elasticities would be extremely large. They are not.

3.9 Short-run versus long-run elasticities

The paper allows for partial portfolio adjustment:

$$\ln B_{it}(h) = \phi \ln B_{it}^A(h) + (1 - \phi) \ln B_{it}^I(h), \quad (15)$$

where B^A is the actively rebalanced allocation and $B_{it}^I(h) = B_{it-1}(h)$ is the inertial component. Short-run elasticities equal ϕ times long-run elasticities.

Interpretation. The baseline demand system mostly captures direct, short-run rebalancing. Long-run responses are larger if investors are sluggish.

3.10 European external-validity regression

At the country level, the paper estimates

$$y_{c,t}(h) - y_{c,t}(s) = \alpha + \beta_0 SIZEIC_c^{2015} \times UFR_t + \beta_1 X_{c,t} + \lambda_t + \nu_c + \varepsilon_{c,t}, \quad (16)$$

where $SIZEIC_c^{2015}$ is the size of the insurance sector relative to debt in country c .

Interpretation. Countries with bigger insurance sectors should see a larger maturity-specific effect once Solvency II makes the UFR relevant across Europe.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the UFR reform is close to exogenous

The core identifying claim is not that nobody had ever heard of the UFR before July 2012, but that market participants did not know the implementation date or the exact shape of the new curve. In particular, they did not know how much the post-reform curve would load on the 20-year market

rate versus the fixed UFR. The sudden jump in the Dutch 30-20 yield spread around July 2, 2012 is consistent with a meaningful surprise.

Why this matters. Anticipated regulation can contaminate demand estimates. The paper argues that anticipation was limited enough that the maturity-pattern shock is usable for identification.

4.2 Why liability duration is the right exposure measure

The reform changes the liability discounting rule, not institutions' asset menu directly. Liability duration measured in 2011q2 therefore acts as a pre-determined treatment-intensity variable. Institutions with more distant promised cash flows should rationally care more about the change in long-end discounting.

4.3 Why the constraint interaction matters

The triple-difference design is not just a robustness exercise. It helps isolate the regulatory-hedging channel. If the reform only changed economic returns, the weakest-funded institutions would not necessarily react more. The fact that constrained investors move more strongly is exactly what the regulatory-hedging mechanism predicts.

4.4 Why quantity evidence is especially persuasive

Prior literature around this reform mostly emphasized price reactions. Jansen's main identification gain is to bring in quantities. Seeing P&Is sell long bonds and buy 20-year bonds in the same maturity-specific pattern implied by the reform makes the pricing evidence much harder to dismiss as pure noise, generic sentiment, or unrelated macro news.

4.5 Alternative explanations the paper confronts

The paper checks several threats:

- **ECB asset purchases and sample-window choice.** The main portfolio shifts are shown not to be driven by APP or by arbitrary pre/post windows.
- **Valuation effects.** Holdings are measured in nominal terms for the portfolio regressions, so the results are about active rebalancing rather than price revaluation.
- **Cross-market substitution.** Allowing for other safe euro-area bonds leaves the main elasticity conclusions broadly unchanged.
- **Weak-instrument worries.** The first stage is strong, but the paper is transparent that some Cragg-Donald statistics are below 10, so some IV estimates should be interpreted with caution.

4.6 What the Dutch design can and cannot identify

What it identifies well:

- the direction of the maturity-specific demand shock,
- the institutions most exposed to it,
- and the sectors most willing to absorb it.

What it identifies less cleanly:

- cross-elasticities across all safe euro-area markets,
- full derivative-inclusive demand curves before 2017,
- and the exact long-run general-equilibrium response once all investors fully adjust.

4.7 Relation to the literature

The paper sits between preferred-habitat theory and demand-based asset pricing.

- Relative to Greenwood and Vissing-Jorgensen (2018), it adds quantities and an explicit demand-system approach.
- Relative to Koijen and Yogo (2019), it contributes a directly observable instrument for yields, based on a regulatory shock rather than latent mandates.
- Relative to the intermediary asset-pricing literature, it highlights banks as the residual absorbers of shocks created by regulation aimed at another sector.

5 Tables and figures

5.1 Figure 1: Dutch 30-20 year yield spread around the reform

Read:

- The Dutch 30-20 year yield spread jumps sharply around July 2, 2012.
- The spread remains noticeably higher after the reform rather than briefly spiking and mean-reverting.
- Visually, this is exactly the pattern one would expect if demand shifts away from 30-year maturities and toward 20-year maturities.

Economic message. The reform steepens the long end immediately, which is the first clue that the maturity-specific demand channel is economically large.

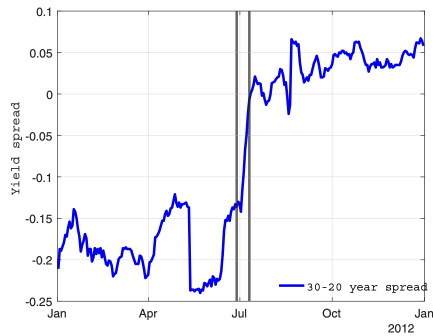


Figure 1
Dutch 30-20 year yield spread: UFR
This graph shows the Dutch 30-20 year government bond yield spread in percentage points around the announcement (and implementation) date of the UFR. The vertical lines represent 5 days before and after the announcement (and implementation) of the UFR on July 2, 2012.

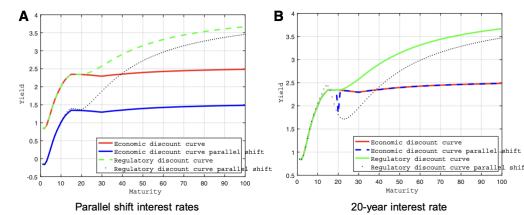


Figure 2
Change in the regulatory discount curve
Panel A shows the economic discount curve (solid red line) and the regulatory discount curve (dashed green line) at implementation of the UFR on July 2, 2012. The graph also shows the economic (solid blue line) and regulatory (dotted black line) discount curve after a parallel shift in market interest rates of $\Delta y_t = -1\%$. Panel B shows the economic discount curve (solid red line) and the regulatory discount curve (solid green line) at implementation of the UFR on July 2, 2012. The graph also shows the economic (dashed blue line) and regulatory (dotted black line) discount curve after a change in the 20-year market interest rate of $\Delta y_t^{(20)} = -0.5\%$.

5.2 Figure 2: How the UFR changes the regulatory discount curve

Read:

- Panel A shows that, relative to the economic curve, the regulatory curve becomes less sensitive to parallel long-rate shifts beyond 20 years.
- Panel B shows that a shock to the 20-year market rate now propagates far out along the regulatory curve.

Economic message. This figure is the institutional heart of the paper. It explains in one picture why regulatory hedging shifts toward 20-year bonds and away from the very long end.

5.3 Table 1 and Figure 3: Sample structure and portfolio rebalancing over time

Read:

- Table 1 shows that pension funds have the longest liabilities and substantial bond holdings; life insurers are also long duration, while nonlife insurers are much shorter duration.
- Figure 3 splits institutions by above- versus below-median liability duration.
- Before the reform, the two groups have fairly similar trends. Right after the reform, the long-duration group cuts 30-year holdings sharply and raises 20-year holdings sharply.

Economic message. The event timing lines up with the exposure measure. This is strong reduced-form evidence before any formal regression is run.

5.4 Table 2: Long-term bond holdings and the regulatory discount curve

Read:

Table 1
Summary statistics

A. AUM							
All assets	Mean	SD	p50	Direct assets	Mean	SD	p50
Life insurers	25,486	20,398	25,903	Life insurers	18,822	15,771	18,208
Nonlife insurers	1,507	1,380	957	Nonlife insurers	1,322	1,359	741
Pension funds	17,578	43,918	6,244	Pension funds	14,761	40,265	3,964
B. Asset allocation							
Government bonds	Mean	SD	p50	Corporate bonds	Mean	SD	p50
Life insurers	49.41	12.53	50.00	Life insurers	33.00	11.70	30.67
Nonlife insurers	45.32	19.75	44.84	Nonlife insurers	34.44	22.95	34.94
Pension funds	37.73	17.20	34.58	Pension funds	20.93	13.33	19.75
Stocks	Mean	SD	p50	Bond duration	Mean	SD	p50
Life insurers	10.08	9.92	6.71	Life insurers	8.72	2.61	8.74
Nonlife insurers	13.75	20.26	3.77	Nonlife insurers	4.47	1.66	4.59
Pension funds	34.13	16.86	36.49	Pension funds	7.29	2.22	7.02
C. Liability information							
Liability duration	Mean	SD	p50	Inverse funding ratio	Mean	SD	p50
Life insurers	11.77	3.95	12.56	Life insurers	91.60	3.42	92.26
Nonlife insurers	4.20	2.62	3.47	Nonlife insurers	91.37	4.01	92.48
Pension funds	17.94	2.96	17.70	Pension funds	92.53	9.43	92.59

This table shows summary statistics on the AUM (panel A), the asset allocation (panel B), and the liability information (panel C) of the P&I sector. In particular, I report the AUM of all assets, AUM of directly reported assets, allocation to government bonds, allocation to corporate bonds, allocation to stocks, bond duration, liability duration, and the solvency positions measured by the funding ratio. The asset allocation and funding ratios are in percentage points, AUM in million euro, bond and liability duration in years. The cross-sectional mean, standard deviation, and median are reported. The quarterly sample period is 2009q1-2019q4.

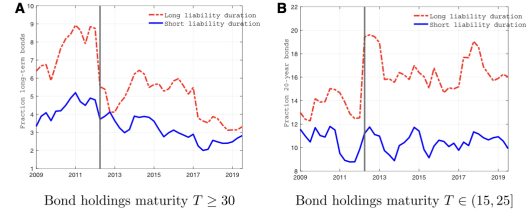


Figure 3

Long-term bond holdings over time

This graph shows the cross-sectional average fraction of the bond portfolio that is invested in 30-year bonds, that is, bonds with a maturity of 30 years or longer (panel A), and for the fraction that is invested in bonds with maturities close to 20 years, that is, between 15 and 25 years (panel B). The sample of P&Is is split in those with long liability durations (above the median) and for those with short liability durations (below the median). The bond portfolio contains the investments in all type of bonds and across all issuer countries. The fractions are in percentage points and the quarterly sample period is 2009q1-2019q4.

- The coefficient on $UFR \times D_{2011q2}^L$ is about -0.164 to -0.157 for 30-year holdings and about $+0.153$ to $+0.244$ for 20-year holdings.
- There is essentially no effect for shorter maturities.
- Economically, the aggregate decline in 30-year holdings is about EUR 10.15 billion, or 33% of pre-reform holdings; the increase in 20-year holdings is about EUR 15.71 billion, or 17%.

Economic message. The maturity pattern is exactly what the model predicts. This is not a generic de-risking or bond selloff; it is a targeted rotation within the long end.

Table 2
Long-term bond holdings and the regulatory discount curve

	Holdings $T \geq 30$		Holdings $15 < T \leq 25$		Holdings $T \leq 15$	
	(1)	(2)	(3)	(4)	(5)	(6)
UFR	0.132 [0.35]		-0.832 [-1.11]		1.109 [0.65]	
D_{2011q2}^L	0.375*** [5.24]		0.267 [1.63]		-0.449* [-1.81]	
$UFR \times D_{2011q2}^L$	-0.164*** [-4.54]	-0.157*** [-5.52]	0.153** [2.41]	0.244*** [5.49]	0.148 [1.11]	0.025 [0.28]
D_{t-1}^L	-0.165*** [-2.62]	0.355*** [4.65]	0.047 [0.29]	0.437** [2.05]	0.473** [2.20]	-1.716*** [-7.43]
FR_{t-1}^{-1}	1.683 [0.59]	3.067 [1.20]	-5.253 [-1.05]	-4.995 [-1.20]	-15.092* [-1.87]	-9.946 [-1.48]
$FR_{t-1}^{-1} \times$ Pension funds	-1.018 [-0.31]	2.935 [0.91]	-3.385 [-0.64]	1.428 [0.29]	7.276 [0.80]	-1.218 [-0.16]
log size	0.880*** [3.65]	-1.38 [-1.19]	0.843*** [2.29]	-1.992 [-1.64]	-2.871*** [-3.98]	6.207** [2.08]
Vol FR_{t-1}^{-1}	-0.133 [-1.52]	-0.099 [-1.40]	-0.349** [-2.12]	-0.195* [-1.79]	0.733*** [2.66]	0.281 [1.47]
Life insurance	4.598*** [9.89]		5.427*** [6.56]		-10.149*** [-6.62]	
Pension funds	2.394 [0.79]		5.108 [1.04]		-16.469** [-2.00]	
Fund FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	2,436	2,436	2,436	2,436	2,436	2,436
Adj R^2	.16	.62	.14	.66	.07	.71

This table presents the results of the main regression described in Equation (6): $w_{it}^B = \alpha + \beta_1 D_{2011q2,t}^L \times UFR_t + \dots + \epsilon_{it}$, with the dependent variable equal to the fraction of the P&I's bond portfolio invested in a certain maturity bucket, UFR equal to one as of 2012q3 and zero otherwise, D_{2011q2}^L the duration of the liabilities as of 2011q2, and controls include the lagged liability duration, the lagged inverse of the funding ratio, the lagged inverse of the funding ratio interacted with a dummy that indicates pension funds, the logarithm of size (AUM), and the annual volatility of the inverse of the funding ratio. Columns 1 and 2 show the results for bond holdings with a maturity of 30 years or longer; columns 3 and 4 for maturities between 15 and 25 years; and columns 5 and 6 for maturities shorter than 15 years. The bond holdings contain the investments in all type of bonds (e.g., corporate, government) and across all issuer countries. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered at the investor level and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 3
Asset allocation and the regulatory discount curve

	Equity	Corporate bonds		Government bonds		
	Allocation (1)	Allocation (2)	DTD (3)	Allocation (4)	Credit rating (5)	High-yield (6)
$UFR \times D_{2011q2}^L$	0.229** [2.37]	-0.176* [-1.96]	-0.015 [-0.97]	-0.059 [-0.59]	-0.038** [-2.00]	0.250** [2.56]
D_{t-1}^L	0.053 [0.30]	-0.009 [-0.04]	-0.018 [-0.52]	-0.067 [-0.30]	0.032 [0.84]	0.433*** [4.05]
FR_{t-1}^{-1}	4.026 [0.67]	22.099*** [3.14]	1.194 [0.99]	-28.305*** [-4.33]	-0.468 [-0.33]	-7.490*** [-1.270]
$FR_{t-1}^{-1} \times$ Pension funds	-9.835 [-1.40]	-7.038 [-0.90]	-5.347*** [-3.74]	21.204*** [2.66]	-4.589*** [-2.94]	1.323 [0.38]
log size	12.852*** [4.35]	-2.891 [-1.11]	1.272*** [2.77]	-3.356 [-1.03]	2.108*** [3.35]	-5.665 [-1.58]
Vol FR_{t-1}^{-1}	0.361** [2.17]	0.142 [0.76]	0.031 [0.90]	-0.576*** [-2.87]	0.039 [1.20]	-0.042 [-0.72]
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,436	2,436	2,436	2,436	2,431	2,431
Adj R^2	.83	.77	.65	.72	.59	.37

This table presents the results of the regression described in Equation (7): $w_{it}^B = \alpha + \beta_1 D_{2011q2,t}^L \times UFR_t + \dots + \epsilon_{it}$, with the dependent variable equal to a measure of P&I's risky asset allocation, UFR equal to one as of 2012q3 and zero otherwise, D_{2011q2}^L the duration of the liabilities as of 2011q2, and controls include the lagged liability duration, the lagged inverse of the funding ratio, the lagged inverse of the funding ratio interacted with a dummy that indicates pension funds, the logarithm of size (AUM), and the annual volatility of the inverse of the funding ratio. Column 1 shows the results for the equity allocation, column 2 for the corporate bond allocation, column 3 for the corporate bond distance-to-default measure (DTD), column 4 for the government bond allocation, column 5 for government bond credit risk, and column 6 for the high-yield government bond allocation. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered at the investor level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 4
Long-term bond holdings and constraints

	Holdings $T \geq 30$		Holdings $15 < T \leq 25$		Holdings $T \leq 15$	
	(1)	(2)	(3)	(4)	(5)	(6)
UFR	-0.491		-3.226*		0.462	
	[-0.40]		[-1.82]		[0.10]	
D_{2011q2}^L	0.509***		0.478**		-0.655*	
	[4.54]		[2.55]		[-1.84]	
FR_{2011q2}^{-1}	-16.576		-51.123***		43.053	
	[-1.55]		[-3.01]		[1.25]	
$D_{2011q2}^L \times FR_{2011q2}^{-1}$	2.335***		2.559**		-5.048**	
	[2.59]		[2.35]		[-2.10]	
$UFR \times D_{2011q2}^L$	-0.201*	-0.280***	0.396***	0.338***	0.154	0.103
	[-1.85]	[-3.92]	[3.22]	[3.71]	[0.49]	[0.56]
$UFR \times FR_{2011q2}^{-1}$	-5	11.912	-23.211	5.563	-3.133	-33.331
	[-0.44]	[1.37]	[-1.21]	[0.42]	[-0.08]	[-1.49]
$UFR \times D_{2011q2}^L \times FR_{2011q2}^{-1}$	-0.444	-1.251**	2.490**	1.2528**	-0.06	0.732
	[-0.47]	[-2.15]	[1.99]	[1.98]	[-0.02]	[0.53]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
P&I type FE	Yes	No	Yes	No	Yes	No
Fund FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	2,432	2,432	2,432	2,432	2,432	2,432
Adj R^2	.18	.62	.16	.66	.09	.71

This table present the results of the regression described in Equation (8): $w_{it}^B = \alpha + \beta_1 D_{2011q2,i}^L \times FR_{2011q2,i}^{-1} \times UFR_i + \dots + \epsilon_{it}$, with the dependent variable equal to the fraction of the P&I's bond portfolio invested in a certain maturity bucket, UFR equal to one as of 2012q3 and zero otherwise, D_{2011q2}^L the duration of the liabilities as of 2011q2, and FR_{2011q2}^{-1} the inverse of the funding ratio minus 1 as of 2011q2. Controls include the lagged liability duration, the lagged inverse of the funding ratio, the lagged inverse of the funding ratio interacted with a dummy that indicates pension funds, the logarithm of size (AUM), and the annual volatility of the inverse of the funding ratio. P&I type fixed effects include dummies indicating pension funds, life insurers, or nonlife insurers. Columns 1 and 2 show the results for bond holdings with a maturity of 30 years or longer, columns 3 and 4 for maturities between 15 and 25 years, and columns 5 and 6 for maturities shorter than 15 years. The bond holdings contain the investments in all type of bonds (e.g., corporate, government) and across all issuer countries. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered at the investor level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$

5.5 Table 3: Asset allocation and the risky-asset channel

Read:

- The interaction $UFR \times D_{2011q2}^L$ raises equity allocation (coefficient about 0.229), reduces corporate-bond allocation (about -0.176), and raises high-yield government-bond exposure (about 0.250).
- A one-standard-deviation increase in liability duration increases equity allocation by roughly 1.49 percentage points and high-yield government-bond exposure by roughly 1.62 percentage points.

Economic message. The reform did not only change duration hedging. It also loosened measured solvency enough to push exposed institutions into riskier assets.

5.6 Table 4: Constraints amplify the maturity shift

Read:

- The triple interaction $UFR \times D_{2011q2}^L \times FR_{2011q2}^{-1}$ is negative for 30-year holdings and positive for 20-year holdings, especially in the fixed-effects specifications.
- A one-standard-deviation increase in the inverse funding ratio strengthens the decline in 30-year holdings by roughly 1.4 percentage points for the average institution and by up to 2.9 percentage points for the most exposed institutions.

Economic message. Institutions closer to their solvency constraints care more about the regulatory curve than about the economic curve, so they rebalance more aggressively.

5.7 Table 5 and Table 6: Instrument and demand-system estimates

Table 5

Instrument for every maturity bucket

	(1,5]	(5,10]	(10,15]	(15,20]	(20,25]	(25,30]	(30,∞)
$\xi(h)$	0	0	0	-0.41	-0.15	0.58	0.91

This table shows the value of the instrument for each maturity bucket used in the instrumental variable approach. The instrument is constructed as the average weight assigned to the UFR for each maturity bucket, minus the average weight assigned to the 20-year interest rate equally distributed over the (15,20] and (20,25] maturity buckets. An overview of the weights for each separate maturity is given in Table [IA2](#).

Table 6

Demand system: Regulatory reform as instrument

A. First-stage IV					
	Foreign $y_t(m)$	Bank $y_t(m)$	MF $y_t(m)$	P&I excl. NL $y_t(m)$	Other $y_t(m)$
$z_t(m)$	0.2806*** [5.87]	0.3287*** [5.09]	0.2766*** [6.55]	0.2824*** [7.06]	0.2902*** [6.30]
Bond controls	Yes	Yes	Yes	Yes	Yes
Initial holdings	Yes	Yes	Yes	Yes	Yes
Observations	260	213	260	260	245
Adj R^2	.97	.97	.97	.97	.97
Cragg-Donald statistic (F -stat)	20.38	18.00	21.30	23.07	20.37
B. Second-stage IV					
	Foreign holdings	Bank holdings	MF holdings	P&I excl. NL holdings	Other holdings
$y_t(m)$	0.5395* [1.68]	2.3526** [2.05]	0.0184 [0.08]	-0.5533** [-1.98]	0.2419 [0.84]
Duration	-0.3421*** [-3.21]	-0.7946** [-2.34]	0.0179 [0.28]	0.1333** [1.97]	-0.1675 [-0.81]
Convexity	0.0095*** [3.26]	0.0217** [2.37]	-0.0004 [-0.25]	-0.0048** [-2.35]	-0.0031 [-0.52]
Coupon	-0.0950*** [-3.67]	0.0098 [0.15]	-0.1095*** [-5.45]	0.0055 [0.20]	0.0311 [0.76]
AUM outstanding	1.3936*** [20.55]	0.9013*** [3.07]	0.7376*** [11.92]	0.2178*** [4.85]	0.3216** [2.18]
10-year German yield	-0.7984** [-2.28]	-2.9318* [-1.86]	0.1724 [0.65]	0.5812* [1.90]	0.0835 [0.10]
Initial holdings	-0.3312*** [-4.58]	0.6789*** [3.63]	0.2102*** [2.92]	0.4873*** [6.05]	0.2465** [2.29]
Observations	260	213	260	260	245
Adj R^2	.92	.76	.92	.87	.92

This table shows the regression results of the demand system described in Equation (13): $\ln B_{i,t}(h) = \hat{\alpha}_i + \beta_{0i} y_t(h) + \beta'_{1i} x_t(h) + \beta_{2i} y_t^{DE} + \beta_{3i} \ln(B_{2009q1,i}(h)) + \epsilon_{it}(h)$. Panel A shows the first stage of the IV and panel B the second stage. The instrument $z_t(h)$ equals the weights assigned to the UFR for each maturity bucket h interacted with a dummy that equals one after implementation of the UFR. Bond characteristics include the average bond duration, convexity, coupon, and the logarithm of AUM outstanding. The outside asset $O_{i,t}$ is proxied by the 10-year German yield and initial holdings are added as control to capture time-invariant omitted characteristics. The quarterly sample period is 2009q1-2019q4. The standard errors are clustered by quarter, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Read:

- Table 5 shows the maturity-bucket instrument: zero at shorter maturities, negative around 20 years, positive beyond 25 years.
- Table 6 first-stage coefficients lie around 0.28 to 0.33 with Cragg-Donald statistics between 18 and 23 in the Dutch-only setup.
- Interpreting a first-stage coefficient around 0.30: yields in the 16–20 year bucket fall about 12 basis points and in the 21–25 bucket about 5 basis points, while yields in the 26–30 bucket rise about 17 basis points and those above 30 years rise about 27 basis points.
- In the second stage, banks and foreign investors have positively sloped demand in yield space (that is, downward-sloping demand in price space), while the non-Dutch P&I sector is upward sloping.

Economic message. The instrument is strong and economically interpretable. It shifts the yield curve in exactly the maturity pattern implied by the regulatory reform.

5.8 Table 7 and the 30-year pricing back-of-the-envelope

Read:

Table 7

Price elasticity of demand

	Obs	Median	SD	p5	p95	Weights
Banks euro	213	14.95	12.58	8.61	47.99	15
Foreign investors	260	3.53	2.21	2.11	9.35	55
Mutual funds euro	260	1.11	0.11	1.06	1.41	16
P&I euro (non-NL)	260	-2.07	4.21	-14.22	-0.75	10
Other euro	245	1.63	1.23	1.03	4.78	4
Total		4.31				100

This table shows the price elasticity of demand, computed as in Equation (12): $-\frac{\partial q_{it}(h)}{\partial p_{it}(h)} = 1 + 100 \frac{\beta_{it}}{f_t(h)} (1 - w_{it}(h))$ for each investor type, maturity bucket h , and quarter t . The median, standard deviation, 5th percentile, and 95th percentile are provided. The total elasticity is the weighted median elasticity, using the weights of each sector defined in the last column. The quarterly sample period is 2009q1-2019q4.

- Median price elasticities are about: banks 14.95, foreign investors 3.53, mutual funds 1.11, other 1.63, and P&Is outside the Netherlands -2.07.
- The weighted aggregate elasticity is 4.31.
- Since Dutch P&Is sell about 22% of the amount outstanding of 30-year Dutch government bonds, a simple price effect is $22/4.31 \approx 5.11\%$, implying about a 17 basis point increase in 30-year yields.

Economic message. The paper translates quantity movements into economically plausible yield effects and identifies the key shock absorbers.

5.9 Figures 4 and 5: Why banks absorb the demand shock

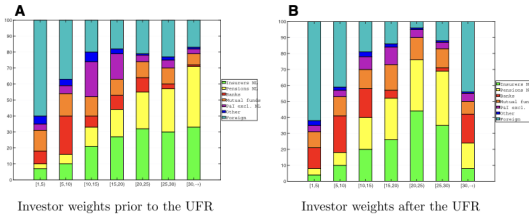


Figure 4
Weights of investor types within maturity buckets

This figure displays the average weights of the investor types (banks, insurance companies, foreign investors, mutual funds, pension funds, and other investors) that held Dutch debt the year prior to the regulatory reform 2011q1-2012q1 (panel A) and the year after the regulatory reform 2012q3-2013q3 (panel B). The banks, mutual funds, and other investor types are at the euro area level. Dutch pension funds, Dutch insurance companies, and the P&I sector in the euro area excluding the Netherlands are separately reported. The foreign sector is determined as the fraction of total amount outstanding that is not held by euro area investors. The weights are in percentage points.

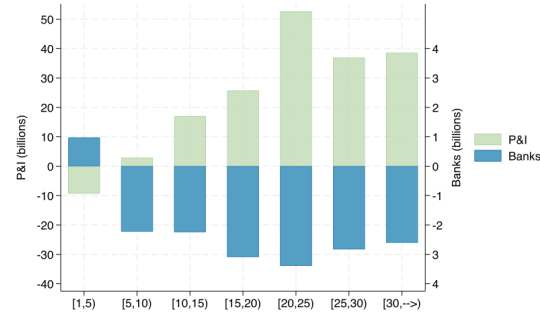


Figure 5
Exposures to interest rates: P&Is versus banks

This figure shows the net exposure of P&Is and banks' swap positions. For the Dutch P&I sector, I compute the aggregate net exposure of their swap positions per maturity bucket. For the Dutch banks, I compute the aggregate net exposures they have with the Dutch P&I sector. The notional amounts to compute the net exposures are based on Euribor plain vanilla swaps. The data come from EMIR and averaged over the period 2018q1-2019q4.

Read:

- Figure 4 shows investor weights in Dutch debt before and after the reform. Banks cut their share of near-20-year bonds from about 9% of total amount outstanding to almost zero, while raising their share of bonds beyond 30 years from about 1% to 18%.
- More than 60% of Dutch bonds beyond 10 years are held by the P&I sector.
- Figure 5 shows that banks hold swap exposures opposite to those of P&Is; EMIR data also show that more than 85% of P&I derivative contracts are with banks.

Economic message. Banks are not passive bystanders. They are connected to P&Is through maturity transformation and swap intermediation, so they naturally become the residual absorbers of the shock.

5.10 Table 8 and Figure 6: European evidence

Table 8

Effects of the UFR at the European level

	Spread 30-20y		Spread 20-10y		Spread 30-10y	
	(1)	(2)	(3)	(4)	(5)	(6)
UFR	0.1278*** [3.34]		0.1879*** [2.60]		0.3157*** [3.56]	
Size IC market	−0.0596*** [−2.62]		0.0744 [1.50]		0.0148 [0.26]	
Size IC market × UFR	0.0511* [1.81]	0.0656*** [2.63]	−0.1043* [−1.71]	−0.1052** [−2.27]	−0.0532 [−0.75]	−0.0396 [−0.71]
10-2y spread	0.0477*** [7.06]	0.0508*** [9.36]	−0.012 [−0.66]	−0.0216 [−0.94]	0.0357 [1.54]	0.0291 [1.05]
Debt to GDP	0.0007 [1.55]	0.0017 [1.60]	0.0031*** [4.00]	0.0027 [1.22]	0.0038*** [3.61]	0.0045 [1.60]
CDS	−0.0005*** [−4.23]	−0.0004*** [−4.09]	−0.0009*** [−3.38]	−0.0009** [−2.57]	−0.0013*** [−3.79]	−0.0013*** [−2.93]
Age	−0.0019 [−0.37]	−0.0336 [−1.50]	0.011 [1.09]	0.0234 [0.82]	0.0091 [0.84]	−0.0102 [−0.32]
Country FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	286	286	286	286	286	286
Adj R^2	.83	.87	.42	.53	.71	.75

This table shows the effects of the UFR on yield spreads for a panel of European countries over the period 2006-2020 (annual). The UFR equals one as of the announcement of the UFR as part of the Solvency II regulation in August 2015, except for the countries Denmark, the Netherlands, and Sweden where the dummy equals one as of June 2012, July 2012, and February 2013, respectively. Size IC market equals the size of the insurance market relative to debt in 2015. Controls include the 10-2y government bond spread, debt-to-GDP ratio, CDS spread, and age of the population. Standard errors are clustered at the country level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

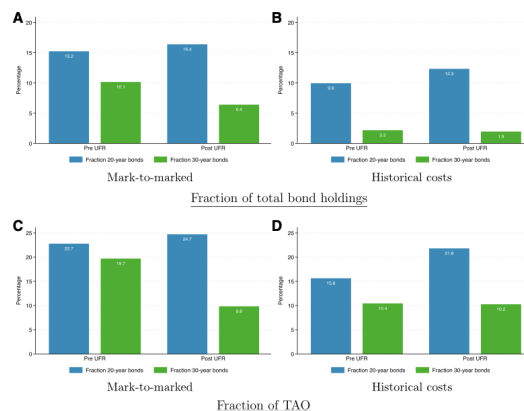


Figure 6

Bond holdings pre/post-implementation of the European-wide UFR

Panels A and B show the average fraction of the bond portfolio invested in 30-year bonds ($T \geq 30$) and the fraction invested in bonds with maturities close to 20 years, that is, $T \in (15, 25]$. The cross-sectional averages are computed both pre- and post-announcement of the European-wide UFR. Panels C and D show the cross-sectional average fraction of the country's bonds outstanding held by the insurance sector in that country, again for the two different maturity buckets. Panels A and C show the results for those countries that applied mark-to-market valuation of the liabilities prior to the UFR (i.e., Ireland, Luxembourg, and the Netherlands) and panels B and D for those that used historical costs valuation (Austria, Belgium, Finland, France, Germany, Greece, Italy, Portugal, Slovakia, and Spain). Finland, Portugal, and Slovakia are excluded from panel D, because for substantial periods of time they are lacking outstanding bonds within either the $T \in (15, 25]$ or $T \geq 30$ maturity bucket. The average allocations pre-UFR (post-UFR) for each country are computed as the time-series average allocation over the period 2009q1-2015q2 (2015q3-2019q4).

Read:

- In the European panel, countries with larger insurance sectors relative to debt experience larger post-UFR increases in the 30-20 spread and larger declines in the 20-10 spread.
- Figure 6 shows a useful split between countries that already used mark-to-market liability valuation and those that used historical costs.
- Mark-to-market countries cut 30-year bonds and raise 20-year bonds after the European-wide UFR. Historical-cost countries mainly increase 20-year holdings, with little reduction in 30-year holdings.

Economic message. The broader European evidence lines up with the Dutch mechanism. Where regulation truly makes liabilities sensitive to market rates, the same maturity rotation appears.

6 Short summary

- The paper exploits a Dutch reform to the regulatory discount curve to identify maturity-specific demand shocks from pension funds and insurers.
- The reform makes liabilities more sensitive to the 20-year rate and less sensitive to very long rates.

- Exposed P&Is cut 30-year bond holdings, buy 20-year bonds, and increase risky asset exposure.
- The long end of the Dutch yield curve steepens as a result.
- Using the reform as an instrument in an asset-demand system, the paper estimates who absorbs the shock.
- Banks are by far the most elastic investors across maturity buckets and are the main absorbers.
- Similar patterns appear in other safe euro-area bond markets and in a Europe-wide insurance-sector exercise.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a causal link from regulation to bond demand and from bond demand to yields. The UFR reform changes the maturity profile of P&I hedging demand, that demand shift shows up clearly in holdings data, and the resulting yield changes can be used to estimate sector-level demand elasticities.

7.2 Economic intuition

The intuition is simple but powerful. Long-term investors want to hedge liabilities. Once regulation says that the relevant liability value depends more on the 20-year rate and less on very long rates, the private incentive is to buy 20-year exposure and sell 30-year exposure. If the investors on the other side of the market are not infinitely elastic, prices must move. Banks matter because they are the main sector willing and able to rearrange maturity exposure quickly.

7.3 Takeaway

This is a paper about the price of regulation in government bond markets. The regulation of one sector, the long-term-investor sector, changes the maturity structure of safe-asset demand and therefore changes the sovereign yield curve. The final lesson is broader than the Dutch case: regulators should think in general-equilibrium terms, because rules aimed at pension funds and insurers can materially reprice long-term government debt and shift risk into banks.